

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 057c46ee637db9b1

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : real_military_patrol_report.iq
File SHA-256 Digest : 057c46ee637db9b18ee414c4bbbbe949f72eac2dfd8b83f15fe4d46bad65a7e1
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:29:49Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	2.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+27.053 kHz	Spectral Centroid Offset
Estimated Baud Rate	250.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	976.6 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	1732.42 kHz	Cumulative PSD 99% Energy
Estimated SNR	10.78 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	2.89 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.1332	Wiener Entropy Index (0.0 – 1.0)

PREDICTED MODULATION : WBFM

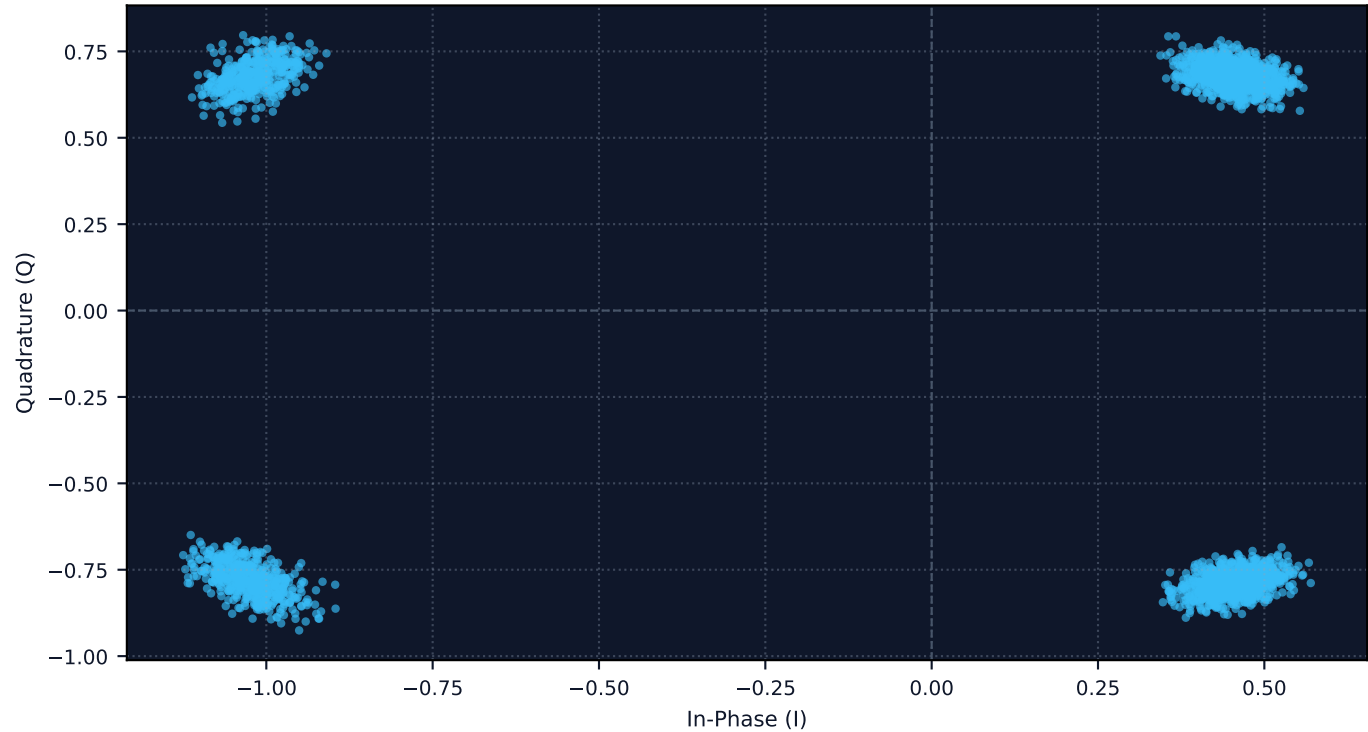
CLASSIFIER CONFIDENCE : 98.77% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: QPSK | Recovered Bits: 16,384 | Estimated EVM: 10.00% | CRC Validated Frames: 0

SIGNAL DISSECTION & DEMODULATION INTELLIGENCE // [WBFM]

Demodulated Baseband Constellation Diagram — [WBFM]



SYNCHRONIZED TELEMTRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	0	0x0000	0	2041 B	CRC MISMATCH	PARTIAL_RECOVER
#2	845	0x0000	1	1935 B	CRC MISMATCH	PARTIAL_RECOVER
#3	888	0x0000	2	1930 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1733	0x0000	3	1824 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1776	0x0000	4	1819 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 84084A482AC9E5A6440A89E4090A2405 [FRAME 1 PAYLOAD]